Disease Simulation Report

Deggendorf Institute of Technology
High Performance Computing and Quantum Computing
Module: HPC Programming Lab

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1 Introduction

Infectious diseases can spread rapidly, often straining healthcare and government systems. Traditional models, based on fixed assumptions like constant transmission rates, struggle to capture the complex, dynamic nature of real-world disease spread.

This project uses a simulation-based SIR model to explore how diseases propagate through a population. By adjusting parameters such as vaccination rate, transmission rate, and disease duration, the simulation reveals trends in infection peaks, susceptibility changes, and the effects of vaccination and recovery over time.

2 Methods

2.1 Model Description

The simulation uses the classical Susceptible-Infected-Recovered (SIR) model. The population is divided into three compartments:

- **Susceptible (S)**: Individuals who can contract the disease.
- **Infected (I)**: Individuals who can transmit the disease.
- **Recovered or Vaccinated (R)**: Individuals who have gained immunity to infection.

2.2 Simulation Logic

The simulation starts by vaccinating a specified percentage of the population. From the remaining population, one individual is marked as infected and the rest as susceptible. Each infected person randomly contacts six individuals per day. If a contacted person is susceptible, they may become infected based on a specified transmission rate. After a defined infectious period, infected individuals recover.

2.3 Parameter Configuration

Key parameters are adjustable via an INI configuration file:

- Population size: 30000
- Initial infected: 1
- Disease duration: 4 days
- Transmission rate (β): 0.25

2.4 Implementation Tools

- **Language**: C++
- **Output**: Daily and statistical S, I, R values stored in CSV files
- **Visualization**: Matplotlib
- **Testing**: Doctest framework
- **Build**: CMake, Docker

3 Results

The simulation demonstrates how an outbreak evolves under specific parameter settings. The data presented is based on the **average of five simulation runs** to ensure consistency and reliability.

3.1 Graphical Trends

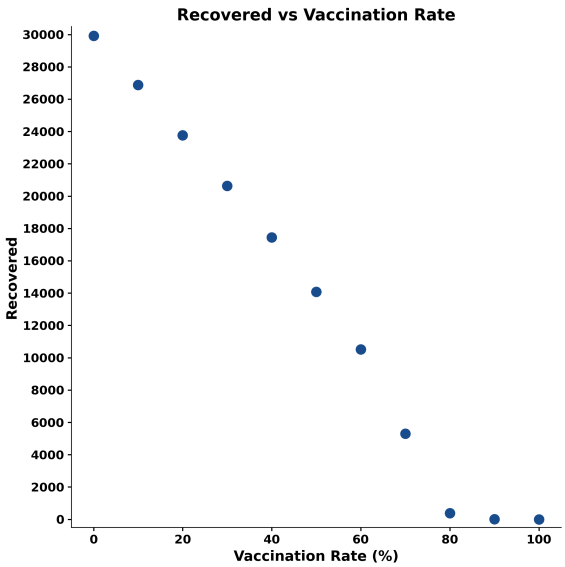

Figure 1: Simulation for D = 4, C = 6, β = 0.25

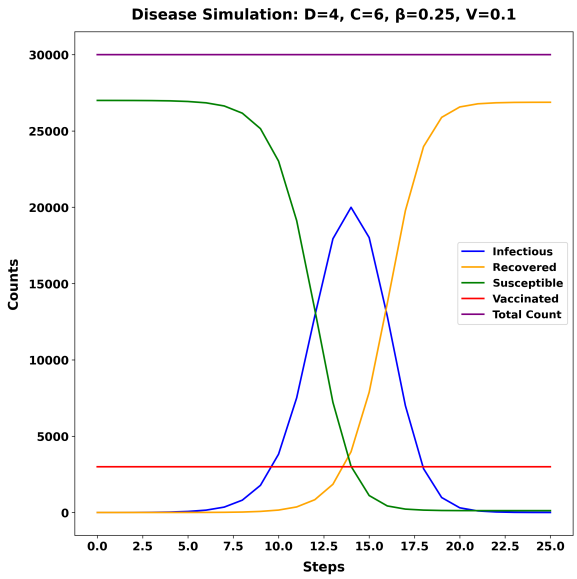

Figure 2: Timestep plot for D = 4, C = 6, β = 0.25 with vaccination rate $V = 0.1$

3.2 Herd Immunity

In the SIR simulation, the basic reproduction number R_0 is calculated as:

$$R_0 = D \cdot \beta \cdot C$$

For this model, where $C = 6$, $\beta = 0.25$, and $D = 4$, we get:

$$R_0 = 4 \cdot 0.25 \cdot 6 = 6.0$$

Based on the Herd Immunity Graph[1], the herd immunity threshold for $R_0 = 6$ is approximately 85%.

3.3 Epidemic Behavior

Based on the graph results, it can be observed that as vaccination rates increase, the number of infected individuals decreases (Figure 1). When about 85% of the population is vaccinated, no more infections occur, showing that herd immunity is reached (Figure 1). Infections rise rapidly and peak around day 14 of the simulation (Figure 2). After that, more people recover each day than getting infected (Figure 2). The outbreak slows down and eventually ends as fewer people are left to be infected with the disease (Figure 2).

4 Discussion

The simulation shows that outbreaks can grow rapidly when the disease spreads easily or has a longer duration. Starting with a portion of the population already vaccinated helps slow the spread by reducing the number of susceptible individuals. While the model simulates the outbreak in a way that is close to real-world scenarios, it has certain limitations and areas for improvement, which are discussed below.

4.1 Limitations

The assumption that everyone randomly interacts with six people suggests an equal contact probability, which may not accurately reflect real-world scenarios. Additionally, vaccination is applied only at the beginning, which is not realistic. The lack of interventions such as lockdowns or isolation further reduces the realism of the simulation. Important factors like age, geographic location are also not included, which could affect the accuracy of the simulation results.

4.2 Areas for Improvement

Considering the limitations, the simulation can be improved by incorporating virus mutations and their effects on transmission and recovery. Regular vaccination schedules can be added to better reflect real-world scenarios. Furthermore, contact patterns can be refined using real-life social network structures. Additionally, interventions such as lockdowns and isolation, along with dynamic factors like age and geographic location, can be included to make the simulation more accurate.

5 Conclusion

In conclusion, this project used a basic simulation to show how diseases spread and how vaccination helps control them. When fewer people are vaccinated, infections grow quickly at first but then go down as people recover. The results show that different vaccination levels lead to different infection peaks—higher vaccination rates cause fewer people to get sick. When about 85% of the population is vaccinated, herd immunity is reached, and new infections stop. This shows that vaccination is a strong and effective way to control and prevent the spread of disease.

However, the simulation has limitations, such as assuming equal contact patterns and applying vaccination only at the start. It also lacks real-world factors like lockdowns, age, location, and virus mutations. Including these could make future models more realistic and accurate.

[1] https://commons.wikimedia.org/wiki/File:Herd_immunity_threshold_vs_r0.svg